Ramlakhan Madheshiya

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SUMMARY

Software engineer with a B.Tech in Computer Science and Engineering (2026). I contribute to the Julia scientific-computing ecosystem, working on the delay differential equation solvers in SciML's OrdinaryDiffEq.jl. I also design and release Godot games and offline-first web applications, and have industry experience in .NET, Python and full-stack JavaScript, including an onsite internship at NTRO.

OPEN-SOURCE CONTRIBUTIONS

SciML / OrdinaryDiffEq.jl Oct 2026 – present

Julia library of ODE, DDE and DAE solvers · 7 pull requests (2 accepted, 5 under review) and 5 issues reported

- Fixed delay differential equation initialization so that zero constant lags are accepted and lags with the wrong sign are rejected, including for reversed time spans and Unitful quantities (#4849).
- Restored the method order of `default_linear_interpolation` so that solver-specific settings apply again, while DAE solvers keep dense output (#4834).
- Fixed the `MethodOfSteps` delay solver for constrained step sizes, SDE caches, composite algorithms and stiff history caches (#4874, #4879, #4892, #4899; under review).
- Enabled discontinuity detection with Unitful and Measurements time types (#4872; under review).
- Reported and diagnosed bugs in the delay and nonlinear solvers: #4875, #4876, #4891, #4895 and NonlinearSolve.jl #1372.

ibis-project / ibis Sep 2024

Python library for portable dataframes

- Improved the documentation site's navigation by making dropdown menus scrollable (#10090).

EXPERIENCE

SDE Intern · National Technical Research Organisation (NTRO) Apr 2025 – Nov 2025

Onsite · three months, in phases

- Designed and built a security tool that supports internal compliance and operations (details confidential under NDA).

Web Developer Intern · Krtya Technologies Pvt. Ltd Jul 2024 – Dec 2024

React, Node.js, Express, PostgreSQL

- Built a full-stack checkout portal with secure payments and order management.

Technical Content Writer · GeeksforGeeks Jan 2024 – Jan 2025

Remote

- Wrote more than 12 technical articles on Python, web technologies, PyTorch and software testing. View articles.

PROJECTS

Brainnova Studio: Games Website · GitHub

GDScript, Godot 4, Android, Web, GitHub Actions

Designed and developed six puzzle and arcade games: Single Shot, Ringburst, Word Glow, Snug Fit, Mirrorlock and Corral. Single Shot (360 rooms) is in closed testing on Google Play. Automated solvers

and tests verify every level, and GitHub Actions builds the web and Android releases.

Bill Studio Website · GitHub

JavaScript, PWA, IndexedDB, Web Crypto

Built an installable billing application that works fully offline. It supports tax rules for more than 50 countries, a drag-and-drop bill designer, customer ledgers, reports, payment QR codes and encrypted backups. All data stays on the user's device.

Group Policy Object Manager for Windows GitHub

C#, .NET, WPF · Smart India Hackathon 2024 finalist

Developed a Windows application that generates and manages Group Policy Objects aligned with CIS security benchmarks, for the NTRO problem statement.

Windows CVE Analysis: Vulnerabilities, Exploits, and Defenses Paper

First-author manuscript · 2025 · with Santosh Saha

Analysis of Windows vulnerabilities, their exploits and the defenses against them.

EDUCATION

Bachelor of Technology, Computer Science and Engineering 2022 – 2026

Uka Tarsadia University, Bardoli, Gujarat, India · CGPA 8.97 / 10

Higher Secondary Certificate (Class 12), Mathematics stream 2022

Shree Vasishtha Vidhyalaya · 91.2%

Secondary School Certificate (Class 10) 2020

Shree Vasishtha Vidhyalaya · 86.4%

ACHIEVEMENTS

- First prize, International Mathematics Olympiad (Class 9).
- Finalist, Smart India Hackathon 2024: reached the final round of India's largest national hackathon with a security tool for NTRO.

SKILLS

- **Languages:** Julia, Python, C, C++, C#, JavaScript, TypeScript, GDScript, HTML/CSS, Shell
- **Frameworks and tools:** Godot, Astro, .NET/WPF, React, Node.js, Flutter, Docker, Git, GitHub Actions

EXTRA-CURRICULAR INTERESTS

- **Gaming:** GTA series, Red Dead Redemption 2, The Witcher 3, Sudoku
- **Cinema:** Films by Christopher Nolan, David Fincher
- **Physical activity:** Football, motorbike riding, cricket